

# **CSE402: Optimisation**

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# Contents

|          |                                                 |          |
|----------|-------------------------------------------------|----------|
| <b>1</b> | <b>Lecture 01</b>                               | <b>4</b> |
| 1.1      | Review on linear algebra . . . . .              | 4        |
| 1.1.1    | Vectors . . . . .                               | 4        |
| 1.1.2    | Linear transformations . . . . .                | 6        |
| 1.1.3    | Basis and Dimension of $\mathbb{R}^n$ . . . . . | 6        |
| 1.1.4    | Dimension . . . . .                             | 10       |
| 1.1.5    | Basis and Dimension of $\mathbb{R}^n$ . . . . . | 11       |
| 1.1.6    | Eigenvectors and eigenvalues . . . . .          | 11       |

# 1 Lecture 01

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Mathematical language of this course is linear algebra (of real vector spaces) and multivariable calculus. In this lecture, we will give a brief review of linear algebra.

## §1.1 Review on linear algebra

**Notation.** Let  $a_1, a_2, \dots, a_n$  be a finite sequence of real numbers. We will denote the sum

$$S = a_1 + a_2 + \dots + a_n \quad (1.1)$$

by

$$S = \sum_{k=1}^n a_k. \quad (1.2)$$

We have the following useful identities:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad (1.3)$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}. \quad (1.4)$$

### §1.1.1 Vectors

**Definition 1.1.1** ( $\mathbb{R}$ -vector space). Let  $V$  be a non-empty set. The set  $V$  equipped with two binary operations

$$+ : V \times V \rightarrow V$$

called *vector addition*, and

$$\cdot : \mathbb{R} \times V \rightarrow V$$

called *scalar multiplication* is called a **vector space over  $\mathbb{R}$**  or  **$\mathbb{R}$ -vector space** if for all  $u, v, w \in V$  and all  $a, b \in \mathbb{F}$ , the following axioms hold:

1. **Closure under addition:**

$$u + v \in V \quad (1.5)$$

2. **Commutativity of addition:**

$$u + v = v + u \quad (1.6)$$

3. **Associativity of addition:**

$$(u + v) + w = u + (v + w) \quad (1.7)$$

4. **Existence of zero vector:** There exists a vector  $0 \in V$  such that

$$u + 0 = u \quad (1.8)$$

5. **Existence of additive inverse:** For every  $u \in V$ , there exists  $-u \in V$  such that

$$u + (-u) = 0 \quad (1.9)$$

6. **Closure under scalar multiplication:**

$$au \in V \quad (1.10)$$

7. **Distributivity of scalar multiplication over vector addition:**

$$a(u + v) = au + av \quad (1.11)$$

8. **Distributivity of scalar multiplication over field addition:**

$$(a + b)u = au + bu \quad (1.12)$$

9. **Associativity of scalar multiplication:**

$$a(bu) = (ab)u \quad (1.13)$$

10. **Identity element of scalar multiplication:**

$$1u = u \quad (1.14)$$

Hence,  $(V, +, \cdot)$  is called a vector space over the field  $\mathbb{R}$ . Elements of  $V$  are called **vectors** and elements of  $\mathbb{R}$  are called **scalars**.

**Convention.** From now on, vector space means  $\mathbb{R}$ -vector space, unless stated otherwise.

In this course, we will work with only the following special example of  $\mathbb{R}$ -vector space.

### Example 1.1.2

Let

$$V := \mathbb{R}^n = \{[x_1, x_2, \dots, x_n]^T \mid x_1, x_2, \dots, x_n \in \mathbb{R}\}. \quad (1.15)$$

For  $[x_1, x_2, \dots, x_n]^T, [y_1, y_2, \dots, y_n]^T \in V$  and  $\alpha \in \mathbb{R}$  define addition (+) and scalar multiplication ( $\cdot$ ), respectively as follows:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad (1.16)$$

$$\alpha \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}. \quad (1.17)$$

Then it is an easy check that  $V = \mathbb{R}^n$  forms an  $\mathbb{R}$ -vector space.

### §1.1.2 Linear transformations

In this course, we are mostly concerned with following special functions between vector spaces:

Not sure about that

**Definition 1.1.3 (Linear transformation).** Let  $V, W$  be two vector space. A **linear transformation** from  $V$  to  $W$  is a function  $T : V \rightarrow W$  such that for all  $\mathbf{x}, \mathbf{y} \in V$  and  $\alpha \in \mathbb{R}$  the following holds true:

- **Additivity.**

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}). \quad (1.18)$$

- **Homogeneity.**

$$T(\alpha \cdot \mathbf{x}) = \alpha \cdot T(\mathbf{x}). \quad (1.19)$$

#### Example 1.1.4

Let  $V := \mathbb{R}^n, W := \mathbb{R}^m$ . An  $m \times n$  matrix  $M$  then defines a linear transformation  $M : \mathbb{R}^n \rightarrow \mathbb{R}^m$ <sup>a</sup> as follows:

For  $\mathbf{x} \in \mathbb{R}^n$ ,

$$M(\mathbf{x}) := M\mathbf{x}. \quad (1.20)$$

Here,  $M$  is an  $m \times n$  matrix,  $\mathbf{x}$  can is an  $n \times 1$  matrix. So  $M\mathbf{x}$  is an  $m \times 1$  matrix. Therefore,  $M$  is indeed a function from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ .

Furthermore, for all  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and  $\alpha \in \mathbb{R}$ , the following holds by the properties of matrix multiplication:

$$M(\mathbf{x} + \mathbf{y}) = M\mathbf{x} + M\mathbf{y}, \quad (1.21)$$

$$M(\alpha\mathbf{x}) = \alpha M\mathbf{x} \quad (1.22)$$

Hence,  $M$  is indeed a linear transformation.

<sup>a</sup>Here, we are denoting the linear transformation determined by the matrix  $M$  by the same letter  $M$  using abuse of notation.

Need to add visualisation of additivity and homogeneity property of a linear transformation.

Surprising fact is that an arbitrary linear transformation also comes from a matrix. Before that, we need the idea of basis.

### §1.1.3 Basis and Dimension of $\mathbb{R}^n$

Consider the vector space

$$\mathbb{R}^2 = \{[x, y]^T : x, y \in \mathbb{R}\}. \quad (1.23)$$

A natural question is: how many vectors are required to generate every vector in  $\mathbb{R}^2$ ? Answer to this question is particularly useful in the context of linear transformations.

Suppose

$$L : \mathbb{R}^2 \rightarrow V$$

is a linear transformation. Because  $\mathbb{R}^2$  contains infinitely many vectors, describing which vector maps to what under  $L$  would require an infinite amount of information.

However, suppose every vector of  $\mathbb{R}^2$  can be generated by a finite collection of vectors

$$v_1, \dots, v_n \quad (1.24)$$

by taking “linear combinations”. Then, instead of knowing the value of  $L$  on every vector, we only need to know the values

$$L(v_1), \dots, L(v_n). \quad (1.25)$$

Indeed, if

$$v = a_1 v_1 + \dots + a_n v_n, \quad (1.26)$$

then by the linearity of  $L$ ,

$$L(v) = L(a_1 v_1 + \dots + a_n v_n) = a_1 L(v_1) + \dots + a_n L(v_n). \quad (1.27)$$

Therefore, knowing the images of  $v_1, \dots, v_n$  completely determines the linear transformation  $L$  on the whole space.

This motivates the following question: What is the smallest collection of vectors required to generate all vectors of a vector space?

The answer leads us to the concepts of spanning sets, linear independence, basis, and dimension:

**Definition 1.1.5 (Linear Independence).** A collection of vectors

$$\{v_1, v_2, \dots, v_k\} \quad (1.28)$$

in a vector space  $V$  is called *linearly independent* if

$$a_1 v_1 + a_2 v_2 + \dots + a_k v_k = 0 \quad (1.29)$$

implies

$$a_1 = a_2 = \dots = a_k = 0. \quad (1.30)$$

If there exists a non-zero solution of the above equation, then the vectors are called *linearly dependent*.

Intuitively, if a list of vectors are linearly dependent, then at least one of them can be generated by rest of them.

#### Example 1.1.6

The vectors

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (1.31)$$

are linearly independent.

However,

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad (1.32)$$

are linearly dependent because the second vector is a scalar multiple of the first.

**Definition 1.1.7 (Span).** Let  $V$  be a vector space and let

$$S = \{v_1, v_2, \dots, v_n\} \quad (1.33)$$

be a collection of vectors in  $V$ . The span of  $S$ , denoted by

$$\text{span}\{v_1, v_2, \dots, v_n\}, \quad (1.34)$$

is the set of all possible linear combinations of the vectors in  $S$ . That is,

$$\text{span}\{v_1, v_2, \dots, v_n\} = \{a_1v_1 + a_2v_2 + \dots + a_nv_n : a_1, \dots, a_n \in \mathbb{R}\}. \quad (1.35)$$

If

$$\text{span}\{v_1, v_2, \dots, v_n\} = V, \quad (1.36)$$

then we say that  $v_1, v_2, \dots, v_n$  span the vector space  $V$ .

A basis is a minimal spanning set, that is, it contains enough vectors to generate the whole space, but no redundant vectors.

### Example 1.1.8

Consider the vectors in  $\mathbb{R}^2$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (1.37)$$

The span of these two vectors is

$$\text{span}\{v_1, v_2\} = \left\{ a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} : a, b \in \mathbb{R} \right\}. \quad (1.38)$$

Now,

$$a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} a+b \\ a-b \end{bmatrix}. \quad (1.39)$$

Given any vector

$$\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2, \quad (1.40)$$

choosing

$$a = \frac{x+y}{2}, \quad b = \frac{x-y}{2} \quad (1.41)$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (1.42)$$

Therefore,

$$\text{span}\{v_1, v_2\} = \mathbb{R}^2. \quad (1.43)$$

Hence, the vectors  $v_1, v_2$  span  $\mathbb{R}^2$ .

**Definition 1.1.9 (Basis).** Let  $V$  be a vector space. A set of vectors

$$\mathcal{B} = \{v_1, v_2, \dots, v_k\} \quad (1.44)$$

is called a *basis* of  $V$  if:

1.  $v_1, v_2, \dots, v_k$  span  $V$ , i.e.

$$V = \text{span}\{v_1, v_2, \dots, v_k\}, \quad (1.45)$$

2.  $v_1, v_2, \dots, v_k$  are linearly independent.

**Example 1.1.10**

Consider the set of vectors

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}. \quad (1.46)$$

We claim that  $\mathcal{B}$  is a basis of  $\mathbb{R}^2$ .

First, let

$$\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \quad (1.47)$$

be an arbitrary vector. We want to express it as a linear combination of the vectors in  $\mathcal{B}$ . Suppose

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (1.48)$$

Then,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a + b \\ a - b \end{bmatrix}. \quad (1.49)$$

Solving the equations

$$a + b = x, \quad a - b = y, \quad (1.50)$$

we get

$$a = \frac{x + y}{2}, \quad b = \frac{x - y}{2}. \quad (1.51)$$

Therefore, every vector in  $\mathbb{R}^2$  can be written as a linear combination of the vectors in  $\mathcal{B}$ . Hence,

$$\text{span}(\mathcal{B}) = \mathbb{R}^2. \quad (1.52)$$

Also, the two vectors in  $\mathcal{B}$  are linearly independent. Therefore,  $\mathcal{B}$  is a basis of  $\mathbb{R}^2$ . Moreover, the representation

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (1.53)$$

is unique. The scalars  $a$  and  $b$  are called the coordinates of the vector with respect to the basis  $\mathcal{B}$ .

### §1.1.4 Dimension

The number of vectors contained in a basis gives the size of a vector space.

#### Theorem 1.1.11

Every basis of a finite-dimensional vector space contains the same number of vectors.

*Proof.* This proof was not shown in the class. Interested students may refer to [Axl24, p. 44].  $\square$

**Definition 1.1.12 (Dimension).** The number of vectors in a basis of a vector space  $V$  is called the *dimension* of  $V$ , denoted by

$$\dim(V).$$

#### Example 1.1.13

Since every basis of  $\mathbb{R}^2$  contains exactly two vectors,

$$\dim(\mathbb{R}^2) = 2.$$

is it exam-  
inable?

### §1.1.5 Basis and Dimension of $\mathbb{R}^n$

The standard basis of  $\mathbb{R}^n$  is

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}. \quad (1.54)$$

Every vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (1.55)$$

can be written as

$$x_1 e_1 + x_2 e_2 + \dots + x_n e_n. \quad (1.56)$$

Therefore,

$$\{e_1, e_2, \dots, e_n\} \quad (1.57)$$

is a basis of  $\mathbb{R}^n$ .

**Theorem 1.1.14**

The dimension of  $\mathbb{R}^n$  is

$$\dim(\mathbb{R}^n) = n. \quad (1.58)$$

**§1.1.6 Eigenvectors and eigenvalues**

Suppose we have a linear transformation  $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ . We are interested in finding a non-zero vector  $\mathbf{x} \in \mathbb{R}^n$  such that

$$L\mathbf{x} = \lambda\mathbf{x}, \quad (1.59)$$

where  $\lambda \in \mathbb{R}$ .

**Definition 1.1.15** (Eigenvector and eigenvalue).  $\mathbf{x}, \lambda$  (if exists) are called an **eigenvector** and **eigenvalue** for  $L$ . The  $(\mathbf{x}, \lambda)$  is called an **eigenpair** for  $L$ .

**Theorem 1.1.16**

Let  $L$  be a linear transformation. Then  $(\mathbf{x}, \lambda)$  is an eigen pair of  $L$  if and only if  $\det(L - \lambda I) = 0$ .

*Proof.* Suppose  $\lambda$  is an eigenvalue of  $L$ . Then there exists a non-zero vector  $\mathbf{x} \in \mathbb{R}^n$  such that

$$L\mathbf{x} = \lambda\mathbf{x}. \quad (1.60)$$

Since  $I\mathbf{x} = \mathbf{x}$ , we have

$$L\mathbf{x} - \lambda I\mathbf{x} = 0, \quad (1.61)$$

or equivalently,

$$(L - \lambda I)\mathbf{x} = 0. \quad (1.62)$$

Since  $\mathbf{x} \neq 0$ , this homogeneous system has a non-trivial solution. Therefore,  $L - \lambda I$  is singular. Hence,

$$\det(L - \lambda I) = 0. \quad (1.63)$$

Conversely, suppose  $\det(L - \lambda I) = 0$ . Then  $L - \lambda I$  is singular, so there exists a non-zero vector  $\mathbf{x} \in \mathbb{R}^n$  satisfying

$$(L - \lambda I)\mathbf{x} = 0. \quad (1.64)$$

Therefore,

$$L\mathbf{x} = \lambda\mathbf{x}. \quad (1.65)$$

Thus,  $\lambda$  is an eigenvalue of  $L$ , with corresponding eigenvector  $\mathbf{x}$ .  $\square$

**Definition 1.1.17** (Eigenvalues of a matrix). Let  $M$  be an  $n \times n$  matrix. The set of all eigenvalues of  $M$  is denoted by

$$\text{eigs}(M) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}. \quad (1.66)$$

Here, the eigenvalues are listed with multiplicity.

**Definition 1.1.18 (Definiteness of a matrix).** Let  $M$  be a real symmetric  $n \times n$  matrix with eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$ . Then  $M$  is called

1. **Positive definite (PD)** if all eigenvalues are strictly positive, i.e.,

$$\lambda_i > 0, \quad \forall i = 1, 2, \dots, n. \quad (1.67)$$

2. **Negative definite (ND)** if all eigenvalues are strictly negative, i.e.,

$$\lambda_i < 0, \quad \forall i = 1, 2, \dots, n. \quad (1.68)$$

3. **Positive semidefinite (PSD)** if all eigenvalues are non-negative, i.e.,

$$\lambda_i \geq 0, \quad \forall i = 1, 2, \dots, n. \quad (1.69)$$

4. **Negative semidefinite (NSD)** if all eigenvalues are non-positive, i.e.,

$$\lambda_i \leq 0, \quad \forall i = 1, 2, \dots, n. \quad (1.70)$$

5. **Indefinite (ID)** if  $M$  has both positive and negative eigenvalues.

### Example 1.1.19

Let

$$L := \begin{bmatrix} -2 & 2 \\ 3 & -1 \end{bmatrix}. \quad (1.71)$$

We will find the eigenvalues and eigenvectors for  $L$ .

The eigenvalues are obtained by solving  $\det(L - \lambda I) = 0$ . Here,

$$\begin{aligned} \det(L - \lambda I) &= \begin{vmatrix} -2 - \lambda & 2 \\ 3 & -1 - \lambda \end{vmatrix} \\ &= (-2 - \lambda)(-1 - \lambda) - 6 \\ &= \lambda^2 + 3\lambda - 4 \\ &= (\lambda - 1)(\lambda + 4). \end{aligned} \quad (1.72)$$

Hence, the eigenvalues for  $L$  are  $\lambda = 1$  and  $\lambda = -4$ .

For each eigenvalue, we find a non-zero vector  $\mathbf{x} = [x_1, x_2]^T$  such that  $(L - \lambda I)\mathbf{x} = \mathbf{0}$ .

For  $\lambda = 1$ , we have

$$\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (1.73)$$

Both equations give  $3x_1 = 2x_2$ , so  $x_2 = \frac{3}{2}x_1$ . Therefore, the eigenvectors corresponding to  $\lambda = 1$  are precisely the vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \frac{3}{2}x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ \frac{3}{2} \end{bmatrix}, \quad x_1 \in \mathbb{R} \setminus \{0\}. \quad (1.74)$$

Choosing  $x_1 = 2$ , we obtain the eigenvector  $\mathbf{x} = [2, 3]^T$ .

For  $\lambda = -4$ , we have

$$\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (1.75)$$

Both equations give  $x_1 + x_2 = 0$ , so  $x_2 = -x_1$ . Therefore, the eigenvectors corresponding to  $\lambda = -4$  are precisely the vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad x_1 \in \mathbb{R} \setminus \{0\}. \quad (1.76)$$

Choosing  $x_1 = 1$ , we obtain the eigenvector  $\mathbf{x} = [1, -1]^T$ . Hence,  $([2, 3]^T, 1)$  and  $([1, -1]^T, -4)$  are eigenpairs for  $L$ .

**Remark 1.1.20.** Any nonzero scalar multiple of an eigenvector is also an eigenvector corresponding to the same eigenvalue.

Indeed, let  $x \neq 0$  be an eigenvector of  $L$  corresponding to the eigenvalue  $\lambda$ . Then

$$Lx = \lambda x. \quad (1.77)$$

For any scalar  $c \neq 0$ , by linearity of  $L$ ,

$$L(cx) = cLx = c\lambda x = \lambda(cx). \quad (1.78)$$

Hence,  $cx$  is also an eigenvector of  $L$  with the same eigenvalue  $\lambda$ .

# References

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